

NARYAN-MAR, Russian Federation

WMO: 232050

Lat: 67.6249N

Lon: 53.0175E

Elev: 39

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.3	-30.7	-41.2	0.5	-36.1	-35.6	0.7	-30.6	21.1	16.0	19.6	15.2	2.3	240	0.783

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	79.0	65.1	73.8	62.3	69.2	59.7	66.9	76.0	64.0	72.0	60.9	67.7	7.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	87.5	71.7	60.4	78.4	68.3	57.4	70.4	65.4	31.5	76.4	29.3	72.5	27.1	67.9	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	17.0	15.3	-40.8	84.1	6.6	5.0	-45.5	87.7	-49.3	90.6	-53.0	93.4	-57.8	97.1		
(5)			-41.0	69.5	6.3	3.4	-45.6	71.9	-49.3	73.9	-52.9	75.8	-57.5	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	1.3	1.4	11.6	22.7	34.1	47.3	56.2	51.1	43.0	31.4	16.2	9.8
	DBStd	22.32	16.82	17.86	14.03	11.32	9.89	9.82	9.27	6.75	6.33	8.64	14.41	16.33
	HDD50	8723	1510	1361	1191	820	505	168	41	66	225	577	1013	1246
	HDD65	13792	1975	1781	1656	1270	959	539	303	433	660	1042	1463	1711
	CDD50	447	0	0	0	0	10	87	234	101	15	0	0	0
	CDD65	42	0	0	0	0	0	8	31	3	0	0	0	0
	CDH74	400	0	0	0	0	3	82	291	24	0	0	0	0
	CDH80	88	0	0	0	0	0	16	69	3	0	0	0	0
	WSAvg	7.4	7.3	7.5	7.7	7.8	7.8	8.0	6.9	6.9	7.1	7.4	7.1	7.6
	PrecAvg	18.5	1.1	0.9	0.9	1.0	1.3	1.7	1.9	2.5	2.2	1.9	1.5	1.3
(15) Precipitation	PrecMax	25.6	2.1	2.4	2.0	2.6	3.4	5.3	4.9	5.4	5.6	3.4	3.9	2.7
	PrecMin	11.1	0.4	0.1	0.2	0.2	0.0	0.5	0.6	0.5	0.2	0.7	0.6	0.5
	PrecStd	3.7	0.4	0.4	0.4	0.6	0.7	1.1	1.1	1.1	1.1	0.6	0.6	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	34.2	38.0	49.1	69.6	82.2	86.0	77.1	65.1	52.2	38.6	35.1
		MCWB	31.7	32.9	34.1	41.4	55.5	63.1	68.8	66.3	57.4	47.9	37.5	33.6
	2%	DB	30.0	31.3	34.8	42.3	61.9	75.3	81.8	71.1	59.7	47.5	35.9	33.6
		MCWB	28.8	30.1	32.5	36.5	51.0	61.5	66.4	62.7	53.8	45.2	34.7	32.5
	5%	DB	27.0	28.3	32.8	38.8	55.1	69.8	76.9	66.7	55.8	44.5	34.4	32.0
		MCWB	25.9	27.1	31.0	35.0	46.7	58.5	64.8	59.8	51.5	43.1	33.3	30.7
	10%	DB	23.6	25.3	30.6	36.3	49.1	64.5	72.2	62.8	52.7	42.0	32.9	29.8
		MCWB	22.5	24.0	28.9	33.7	43.3	55.5	61.8	57.8	49.4	40.5	31.9	28.6
	0.4%	WB	32.0	33.4	34.6	41.6	57.4	66.6	71.5	68.1	59.5	48.6	37.7	34.1
		MCDB	33.0	34.2	36.9	49.1	68.7	79.3	81.6	75.1	63.3	51.2	38.2	34.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.0	30.2	33.0	37.5	51.5	63.1	68.6	63.7	54.7	45.3	35.0	32.7
		MCDB	30.0	31.2	34.7	41.1	60.2	73.0	77.8	69.5	58.6	47.1	35.6	33.5
	5%	WB	25.9	27.2	31.3	35.5	47.1	59.3	66.0	60.7	52.0	43.2	33.5	30.7
		MCDB	26.8	28.3	32.7	37.9	54.1	67.8	74.7	65.8	55.0	44.6	34.2	31.8
	10%	WB	22.6	24.1	28.9	33.9	43.3	55.8	63.1	58.1	49.6	40.7	32.0	28.6
		MCDB	23.6	25.3	30.6	36.1	49.4	64.1	70.7	62.5	52.5	41.9	32.9	29.8
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	13.6	15.1	13.9	12.7	13.8	15.1	11.6	10.5	7.6	10.2	12.1
		MCDBR	11.4	12.5	9.0	14.0	23.2	23.2	22.4	19.5	15.9	7.8	7.7	9.2
		MCWBR	11.2	12.4	8.1	10.3	13.9	11.9	11.0	11.6	10.3	6.7	7.9	8.9
	5% WB	MCDBR	11.7	12.5	8.6	11.7	21.9	21.2	20.9	18.0	14.3	7.6	7.5	9.4
		MCWBR	11.5	12.4	8.1	9.1	13.7	12.2	11.4	12.0	10.2	6.9	7.8	9.2
(40) Clear-Sky Solar Irradiance	taub		0.215	0.259	0.300	0.330	0.344	0.348	0.381	0.359	0.324	0.281	0.213	N/A
	taud		1.903	2.033	2.087	2.122	2.293	2.462	2.375	2.423	2.477	2.232	1.942	N/A
	Ebn at Noon		31	168	229	257	269	272	256	249	231	163	29	0
	Edh at Noon		3	16	28	36	34	30	32	27	20	14	3	0
(44) All-Sky Solar Radiation	RadAvg		12	157	638	1214	1535	1563	1513	974	541	170	30	0
	RadStd		2	23	62	94	140	162	204	96	56	25	3	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page